

Chapter 1.

1.1. (a) $1 - P_0 - P_1 - P_2 - P_3 = 0.05$

(b) $P_0 + P_1 + P_2 = 0.8$

1.2. $C = \{\text{Cloudy}\}$ $R = \{\text{Rainy}\}$

$$P\{CUR\} = P\{C\} + P\{R\} - P\{C \cap R\}$$

$$= 0.4 + 0.3 - 0.2$$

$$= 0.5$$

$$\Rightarrow P\{(CUR)^c\} = 1 - P\{CUR\} = 0.5$$

1.3. (a) $\frac{8}{14} \cdot \frac{7}{13} = \frac{4}{13}$

(b) $\frac{6}{14} \cdot \frac{5}{13} = \frac{15}{91}$

(c) $\frac{8}{14} \cdot \frac{6}{13} + \frac{6}{14} \cdot \frac{8}{13} = \frac{48}{91}$

1.4. (a) $\frac{27}{58}$

(b) $\frac{27}{35}$

1.6. $A = \{\text{they are both aces}\}$

$B = \{\text{they are of different suits}\}$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{4}{52} \cdot \frac{3}{51}}{\frac{39}{51}} = \frac{1}{169}$$

1.8. (a) $S = \{(L, L), (L, W), (W)\}$

$$P\{X > 0\} = P\{(W), (L, W)\} = P\{X = 1\}$$

$$= \frac{18}{38} + \frac{20}{38} \cdot \frac{18}{38} = \frac{261}{361}$$

$$P\{X < 0\} = P\{(L, L)\} = P\{X = 3\}$$

$$= \left(\frac{20}{38}\right)^2 = \frac{100}{361}$$

(b) $E[X] = 1 \cdot \frac{261}{361} + (-3) \cdot \frac{100}{361} = -\frac{39}{361}$

1.10. $P\{N=2\} = 0.5$

$$P\{N=3\} = 0.5$$

(a) $E[N] = 2(0.5) + 3(0.5) = 2.5$

(b) $Var(N) = E[(N - E(N))^2]$

$$= (2 - 2.5)^2 \cdot 0.5 + (3 - 2.5)^2 \cdot 0.5$$

$$= \frac{1}{4}$$

1.11. Proof.

$$\begin{aligned} Var(X) &= E[(X - E[X])^2] \\ &= E[X^2 - 2XE[X] + E[X]^2] \\ &= E[X^2] - 2E[XE[X]] + E[E[X]^2] \\ &= E[X^2] - 2E[X]E[X] + E[X]^2 \\ &= E[X^2] - E[X]^2 \end{aligned}$$

1.12. Let F be her fee if she takes the fixed amount and X when she takes the contingency amount.

$$E[F] = 5000, SD(F) = 0$$

$$E[X] = 25000 \cdot (0.3) + 0 \cdot (0.7) = 7500$$

$$E[X^2] = (25000)^2(0.3) + 0 \cdot (0.7) = 1.875 \times 10^8$$

$$\begin{aligned} \Rightarrow SD(X) &= \sqrt{Var(X)} = \sqrt{E[X^2] - (E[X])^2} \\ &= \sqrt{1.875 \times 10^8 - (7500)^2} \\ &= \sqrt{1.3125} \times 10^4 \end{aligned}$$

1.13. (a) $E[\bar{X}] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n} \cdot (n\mu) = \mu$

(b) $Var(\bar{X}) = Var\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} Var\left(\sum_{i=1}^n X_i\right)$

$$= \frac{1}{n^2} \sum_{i=1}^n Var(X_i) = \frac{1}{n^2} \cdot (n\sigma^2)$$

$$= \frac{\sigma^2}{n}$$

$$\begin{aligned}
 (c) \sum_{i=1}^n (X_i - \bar{X})^2 &= \sum_{i=1}^n (X_i^2 - 2X_i\bar{X} + \bar{X}^2) \\
 &= \sum_{i=1}^n X_i^2 - 2\bar{X} \sum_{i=1}^n X_i + n\bar{X}^2 \\
 &= \sum_{i=1}^n X_i^2 - 2n\bar{X}^2 + n\bar{X}^2 \\
 &= \sum_{i=1}^n X_i^2 - n\bar{X}^2
 \end{aligned}$$

$$\begin{aligned}
 (d) E[(n-1)S^2] &= E\left[\sum_{i=1}^n (X_i - \bar{X})^2\right] \\
 &= \sum_{i=1}^n E[X_i^2] - nE[\bar{X}^2] \\
 &= \sum_{i=1}^n (E[X_i^2] - E[\bar{X}^2]) \\
 &= n(E[X^2] - E[\bar{X}^2]) \\
 &= n(\sigma^2 + \mu^2) - n(\frac{\sigma^2}{n} + \mu^2) \\
 &= (n-1)\sigma^2
 \end{aligned}$$

$$\Rightarrow E[S^2] = \sigma^2$$

$$\begin{aligned}
 1.14. \text{Cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] \\
 &= E[XY - XE[Y] - YE[X] + E[X]E[Y]] \\
 &= E[XY] - 2E[X]E[Y] + E[X]E[Y] \\
 &= E[XY] - E[X]E[Y]
 \end{aligned}$$

$$1.15. (a) \text{Cov}(X, Y) = \text{Cov}(Y, X), \text{ trivial from definition.}$$

$$(b) \text{Cov}(X, X) = E[(X - E[X])^2] = \text{Var}(X)$$

$$\begin{aligned}
 (c) \text{Cov}(cX, Y) &= E[(cX - E[cX])(Y - E[Y])] \\
 &= cE[(X - E[X])(Y - E[Y])] \\
 &= c\text{Cov}(X, Y)
 \end{aligned}$$

$$\begin{aligned}
 (d) \text{Cov}(c, Y) &= E[(c - E[c])(Y - E[Y])] \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 1.16. \text{Cov}(X, Y) &= \text{Cov}(aU + bV, cU + dV) \\
 &= ac\text{Cov}(U, U) + ad\text{Cov}(U, V) \\
 &\quad + bc\text{Cov}(V, U) + bd\text{Cov}(V, V) \\
 &= ac + bd
 \end{aligned}$$

1.17.

$$\begin{aligned}
 (a) \text{Cov}(X_1 + X_2, X_3 + X_4) \\
 &= \text{Cov}(X_1, X_3) + \text{Cov}(X_1, X_4) + \text{Cov}(X_2, X_3) \\
 &\quad + \text{Cov}(X_2, X_4) \\
 &= 1 \cdot 3 + 1 \cdot 4 + 2 \cdot 3 + 2 \cdot 4 = 21
 \end{aligned}$$

(b) Similarly, we get

$$\text{Cov}(X_1 + X_2 + X_3, X_2 + X_3 + X_4) = 54.$$

1.18. Let X_i be the amount the stock goes up in period i , then.

$$Y = \sum_{i=1}^3 X_i$$

$$\begin{aligned}
 \text{Cov}(X_1, Y) &= \text{Cov}(X_1, X_1 + X_2 + X_3) \\
 &= \text{Cov}(X_1, X_1) + \text{Cov}(X_1, X_2) \\
 &\quad + \text{Cov}(X_1, X_3) \\
 &= \text{Cov}(X_1, X_1) \\
 &= \text{Var}(X_1) = E[(X_1 - E[X_1])^2] \\
 &= E[X_1^2] \quad (\text{since } E[X_1] = 0) \\
 &= 1^2 \cdot 0.5 + (-1)^2 \cdot 0.5 = 1
 \end{aligned}$$

$$\text{Var}(Y) = \sum_{i=1}^3 \text{Var}(X_i) = 1 + 1 + 1 = 3.$$

$$\begin{aligned}
 \Rightarrow \text{Corr}(X_1, Y) &= \frac{\text{Cov}(X_1, Y)}{\sqrt{\text{Var}(X_1)\text{Var}(Y)}} \\
 &= \frac{1}{\sqrt{1 \cdot 3}} = \frac{1}{\sqrt{3}}
 \end{aligned}$$

1.19. No, since if $\text{Var}(X) = \text{Var}(Y) = 1$

$$\begin{aligned}
 |\text{Cov}(X, Y)| &\leq \sqrt{\text{Var}(X)\text{Var}(Y)} \\
 &\leq 1
 \end{aligned}$$